

EVICT, QUANTIZE, OR ALLOCATE?

A PHASE DIAGRAM FOR KV-CACHE COMPRESSION

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ABSTRACT

KV-cache compression has split into two literatures: *quantization* keeps every token at reduced precision, *eviction* keeps few tokens at full precision. We show they are two corners of one rate-allocation problem. Posed against the distortion of the attention *output* rather than of the keys, the optimum is reverse water-filling, $b_i^* = \log_2 a_i + c$, and eviction is its zero-rate tier at a cost $c_0=1$ that is derived rather than tuned. The same cost comparison yields a regime variable: the fraction of heads whose 2-bit tier is already costlier than eviction ($\tau^2 c_2 > c_0$). Across 10,240 heads of six LLMs at 8k–128k context, measured by exact recomputation, this *dead-tier fraction* orders the gain of allocation over the best corner at Spearman $\rho = -0.96$ (-1.00 within every model), contains no context length, and correctly anticipated a held-out collapse at 128k. Context length turns out to be a phase variable: a single model crosses from the allocation-dominated to the eviction-dominated regime as L grows, so “method X beats Y on model M” is ill-posed without a stated L . We turn the map into SIEVE, a per-head router calibrated in one forward pass. **TBD [R8]: end-task result, one sentence.**

1 INTRODUCTION

At long context the KV cache, not the weights, bounds batch size and decode throughput. Two families of methods compete to shrink it. Quantizers (Liu et al., 2024; Hooper et al., 2024; Zandieh et al., 2024; 2025) store every token at 2–4 bits; evictors (Zhang et al., 2023; Liu et al., 2023; Li et al., 2024; Xiao et al., 2024) store a subset of tokens at full precision and drop the rest. Each reports wins over the other, and practitioners pick by leaderboard.

These are not competing ideas; they are the endpoints of one. Attention does not consume keys, it consumes $o = \sum_i a_i v_i$. Once the objective is the error in o , the question “how many bits should token i receive?” has a closed-form answer that ranges over $\{0, \dots, b_{\max}\}$. Uniform quantization is the answer when every token gets the same width; eviction is the answer when only 0 and $b_{\max}$ survive. Neither family is right or wrong in general. The useful question is *when* each one is optimal, and whether that can be read off the model before deploying anything.

Our answer is a phase diagram (Fig. 1). The optimal bit profile has three shapes (Fig. 1a): flat when attention is diffuse, a staircase (the *productive band*, where mixed precision beats both corners), and keep/evict when attention is sharp. The boundary between the last two is not an empirical threshold. It follows from the same inequality that produces the allocation: a precision tier is worth buying only while its logit noise $\tau^2 c_b$ costs less than evicting the token ($c_0 = 1$). The fraction of heads for which the 2-bit tier fails this test is our order parameter, and Fig. 1b shows 24 measured configurations collapsing onto it.

Contributions.

1. **One objective for both literatures** (§2). Under output distortion, eviction is the zero-rate tier of a water-filling allocator with a derived cost $c_0 = 1$, and the optimal rule $b_i^* = \log_2 a_i + c$ reads only attention weights.
2. **A derived order parameter** (§3). The dead-tier fraction predicts where allocation pays across six architectures and five octaves of context ($\rho = -0.96$ pooled, -1.00 within each model). It contains no L , holds when the eviction baseline is redefined, and passed a held-out test.

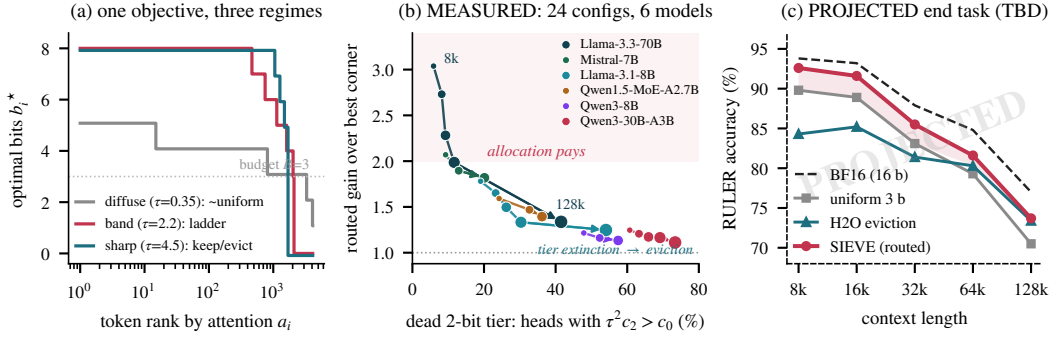


Figure 1: **(a)** Optimal bit-widths from reverse water-filling at a 3-bit average budget (synthetic Gaussian logits, measured cost constants). As logit spread τ grows the profile goes from near-uniform to a ladder to keep/evict. **(b)** *Measured*: routed gain of allocation over the better of uniform quantization and eviction, per (model, context); marker size grows with context and arrows mark the last doubling. One derived variable orders all 24 points ($\rho = -0.96$), and every model moves right as context grows. **(c)** *Projected, not measured (TBD [R8])*: the end-task picture the phase diagram predicts, with the largest gain at short-to-mid context, shrinking toward eviction at 128k.

- Context length is a phase variable.** τ rises monotonically with L in every model. Rising τ kills precision tiers from the bottom, so one model crosses regimes as context grows and a per-model verdict is ill-posed.
- SIEVE, a router built from the map (§4).** One calibration pass assigns each head to uniform, ladder, or eviction. It rests on a cost model that forward-predicts measured gains to within 0.79–1.34 $\times$, and on measured design facts (Table 1).

2 ONE OBJECTIVE: EVICTION IS QUANTIZATION AT ZERO RATE

Output distortion. For query q and keys k_i , let $s_i = q^\top k_i / \sqrt{d}$, $a = \text{softmax}(s)$, $o = \sum_i a_i v_i$. Storing k_i at b_i bits perturbs the logit by δ_i with $\text{Var}(\delta_i) = \tau^2 c_{b_i}$. Here τ is the head’s logit spread and c_b the quantizer’s relative distortion; we use the variance because softmax is shift-invariant, so bias costs nothing. Since $\partial o / \partial s_i = a_i(v_i - o)$, to first order

$$\mathbb{E} \|\Delta o\|^2 \approx \sum_i a_i^2 \|v_i - o\|^2 \tau^2 c_{b_i}. \quad (1)$$

Evicting a set E and renormalizing costs $\sum_{i \in E} a_i^2 \|v_i - o\|^2$ to the same order. That is Eq. 1 with $\tau^2 c_0 = 1$, so eviction is a bit-width like any other and **its cost constant is derived, not chosen**.

Reverse water-filling. Minimizing Eq. 1 subject to $\sum_i b_i \leq BL$ is a classical rate-allocation problem (Cover & Thomas, 2006). The Lagrangian separates per token, and with $c_b \propto 4^{-b}$ its solution above the water level is $b_i^* = \log_2 a_i + \log_2 \|v_i - o\| + c$, with $b_i^* = 0$ (eviction) below it. We measure the value term at ≤ 0.035 bits everywhere in our data, so in practice

$$b_i^* = \log_2 a_i + c, \quad (2)$$

which reads attention weights only and never touches the value cache.

Tier extinction and the order parameter. A tier b lies on the lower convex envelope of the cost curve only if storing a token at b bits beats dropping it: $\tau^2 c_b < c_0 = 1$. When $\tau^2 c_b > 1$ the tier is *dead*, meaning no token should ever be stored at b bits. This one inequality produces both edges of the band. At the *sharp* end, rising τ kills rungs from the bottom until only $\{0, b_{\max}\}$ remain, and a two-tier allocation *is* eviction. At the *diffuse* end, the ladder’s spread, $\text{std}_i(\log_2 a_i) = \tau / \ln 2$, drops below about one rung and the allocation coincides with uniform. We define the **dead-tier fraction** D_2 of a (model, context) configuration as the share of heads with $\tau^2 c_2 > 1$. It uses one calibration pass, has no free parameter, and contains no L .

3 MEASURING THE PHASE DIAGRAM

Setup. We use six models: Llama-3.1-8B and Llama-3.3-70B (Grattafiori et al., 2024), Mistral-7B (Jiang et al., 2023), Qwen1.5-MoE-A2.7B, Qwen3-8B, and Qwen3-30B-A3B (Yang et al.,

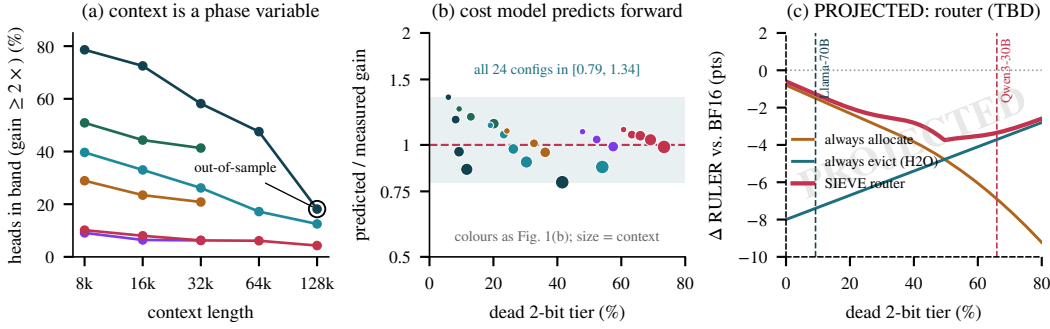


Figure 2: **(a)** Share of heads where allocation beats the best corner by $\geq 2\times$, versus context (colours as Fig. 1b). Every model decays, at -1.4 to -14.6 points per doubling; the circled point was held out. **(b)** The linearized cost model’s predicted gain over uniform divided by the exactly recomputed gain: all 24 configurations fall within $[0.79, 1.34]$, while the measured gains themselves span $20\text{--}37\times$. **(c)** *Projected, not measured* (TBD [R8]): the hypothesis the router experiment tests. Always-allocate wins at low D_2 and always-evict at high D_2 ; a per-head router should track the upper envelope, with the largest margin in the mixed middle.

2025). Each runs at every power-of-two context from 8k up to its native window, giving 24 configurations and 46k head measurements. Prompts are PG-19 haystacks (Rae et al., 2020) with an embedded needle. Every configuration passes an input-validity gate (real text; needle retrieved on every prompt). Keys are quantized with TurboQuant_{mse} (Zandieh et al., 2025) (random rotation plus Lloyd–Max (Lloyd, 1982)), and every reported error is an *exact recomputation* of o with the real quantized keys. Eq. 1 only chooses bit-widths and is never used to score them. At a matched budget of $B=3$ bits/token we compare the water-filling allocation against two corners: uniform B -bit quantization, and eviction keeping $BL/8$ tokens at 8 bits, ranked by H2O accumulated attention (SnapKV/StreamingLLM rankings win $\leq 22\%$ of heads). A head’s *gain* is the best corner’s error divided by the allocation’s error. A head is *in band* if its gain is at least $2\times$, and *routed gain* is the geometric mean of $\max(\text{gain}, 1)$, which is what a per-head router can realize. Our primary comparison is *symmetric*: allocation and corner score tokens with the same information. TBD [R3]: a fully deployable symmetric cell (both sides on lagged attention) agrees with this one to within 5 band-points on 4 configurations; the re-run with the fixed instrumentation is in flight. Add seeds and error bars on the 128k cells [R7].

Finding 1: the dead-tier fraction is the order parameter. Across 24 configurations D_2 orders routed gain at Spearman $\rho = -0.963$ (Fig. 1b). Configurations nested in six models are not independent points, so we also partial out model identity, which leaves $\rho = -0.944$; within *every* model the relationship is perfectly monotone ($\rho = -1.00$). The obvious alternatives do worse. Logit spread τ alone gives -0.76 pooled, because it does not know where the tiers fall relative to c_0 . Normalized effective support n_{95}/L , a natural “concentration” index, gives $+0.00$, because dividing by L confounds architecture with the context at which it was measured. The order parameter also survives a change of target: replacing the corner’s ranking by a deployable H2O score moves band fractions by 6–29 points yet leaves $\rho = -0.941$. Finally, it passed a held-out test. The boundary was fit without Llama-3.3-70B at 128k, and at that point D_2 jumps from 11.7% to 41.5% and the band falls from 47.5% to 18.1%, as the fitted relation predicts (Fig. 2a, circled).

Finding 2: context length is a phase variable. Within every model τ rises monotonically with L (rank correlation 1.00 in 6/6). Rising τ pushes $\tau^2 c_b$ past c_0 for successively higher rungs, the ladder is eaten from the bottom, and the optimum degenerates toward eviction. In absolute terms allocation keeps improving over uniform with context ($26\times \rightarrow 34\times$ on Llama-3.1-8B); what changes is that the eviction corner catches up. Llama-3.3-70B is the most allocation-friendly model in the study at 8k–64k (78.6% of heads in band at 8k) and among the least at 128k (Fig. 2a). *Architecture rankings are not stable across context*, so a single verdict per model, the way this literature reports results, is not well defined. The same corner can also be reached for a different reason. Qwen3 models sit at 60–73% dead tiers at *every* length because of sink-dominated attention, which is architectural, whereas Llama-3.1-8B reaches the same corner by traversal. TBD [R4]: attention or RoPE limit? The collapsing Llamas sit at their trained window; Qwen3-30B, at 50% of its window, drops least. Planned: 96k Llamas; Qwen3-30B at 192k/256k.

Table 1: Measured facts (24 configurations) and the design decision each one forces.

Measurement	Consequence for SIEVE
Water-filling evicts 40–54% of tokens by itself at $B=3$	Eviction is tier 0 <i>inside</i> the allocator
1-bit tier dead in 95–100% of heads	Never offer 1 bit
Allocation beats uniform 3-bit by 20–37×	The baseline to beat is eviction, not uniform
Oracle token ranking beats H2O by only 1.02–1.41×	Scoring is solved; the value is in <i>allocation</i>
K^*/n_{95} spans 5.8–215, not ordered by support n_{95}	No global keep-count constant; derive per head
Ladder width vs. realized gain: $\rho \in [-0.77, -0.41]$ in 24/24	Cheap per-head routing signal, no quantization

Finding 3: the cost model predicts forward. The constant $\tau/\ln 2$ ladder width is an algebraic identity, not evidence. The real test of Eq. 1 is whether it predicts *measured* error ratios. It does: per-configuration median predicted/measured gain lies in $[0.79, 1.34]$ across all 24 configurations (Fig. 2b), for a quantity that spans 20–37×

4 FROM MAP TO METHOD: SIEVE

Any single policy is wrong for some heads at some context, so SIEVE routes per head and per deployment length.

Offline calibration (one forward pass). For each head, measure τ and the per-tier distortion c_b , which gives the set of live tiers and the ladder width. Route each head to *uniform* when its ladder is narrower than about one rung, to *eviction* (H2O) when the tiers below $b_{\max}$ are extinct, and to the *water-filling ladder* otherwise. The ladder’s tier set is read off the convex envelope: tier 0 included, 1-bit excluded. Because τ is monotone in L and D_2 is L -free, a head is re-phased for a new deployment length by re-evaluating the same inequality, with no new measurement. **TBD [R6]: pin the boundary as an interval ($D_2 \in [40, 60]\%$).**

Decode. Band heads run a cascade. A 3-bit base pass over all tokens yields $\hat{a}$; Eq. 2 allocates $b_i^* = \log_2 \hat{a}_i + c$; successively refinable codes (Equitz & Cover, 1991) add $+1/ + 2/ + 2$ bits on shrinking sets; the tail is evicted. **TBD [R3/R5]: re-budgeting cadence.** Preliminary: stale scores cost in-band heads 1.4–1.6×

but out-of-band heads only 1.00–1.03×, so decode-time re-budgeting may be load-bearing. **TBD [R11/R12/R14]: nested-code overhead vs. monolithic 8-bit (kill at ~10%); GQA union cost; kernel TPOT vs. dense FP8 at 128k.**

End-task validation. **TBD [R8]:** Router on versus off on RULER (Hsieh et al., 2024), at matched bits, on the highest- and lowest- D_2 models (Qwen3-30B-A3B; Llama-3.3-70B at $\leq 64k$). The falsifiable prediction (Fig. 2c) is that the router recovers accuracy *exactly where the diagram says the allocator degenerates*, and that the end-task gap shrinks with context as in Fig. 1c. **TBD [R9]: K^* -derived per-head budgets vs. Feng et al. (2024); Cai et al. (2024) on LongBench (stands alone).**

5 RELATED WORK AND OUTLOOK

KV quantization ranges from per-channel/per-token integer schemes (Liu et al., 2024; Hooper et al., 2024) to rotation-based and sketching quantizers (Zandieh et al., 2025; 2024). **Mixed precision** (Yang et al., 2024; He et al., 2024) gives salient tokens heuristic higher tiers. **Eviction** scores tokens by accumulated or windowed attention (Zhang et al., 2023; Li et al., 2024; Liu et al., 2023), keeps sinks (Xiao et al., 2024), or reallocates budget across layers and heads (Cai et al., 2024; Feng et al., 2024). Each optimizes *within* a family; we derive the boundary *between* families. Tier extinction also explains why QJL’s unbiased 1-bit tier is never worth offering: it costs more than eviction. We study the *token* axis of the key cache; per-channel key treatment is orthogonal and composes with ours. **TBD [R15]: related-work sweep (last 12 months) for concurrent output-distortion and rate-allocation formulations; position against them explicitly.**

Limitations. Every number here is an output-error ratio; whether a 1.3–3× routed gain survives to end-task accuracy and throughput is the main open question (**TBD [R8], [R14]**), and reach beyond 128k hinges on [R4]. The diagram does not depend on either: it says where to spend bits before deployment.

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A APPENDIX (PLANNED)

TBD A.1 Measurement integrity: the validity gates (real-corpus enforcement, needle retrieval, probe/chunking equivalence) and the defects they catch. A.2 The $\tau/\ln 2$ ladder identity as a consistency check; the 2-bit quantizer acts as a universal logit temperature ($\alpha = 0.941\text{--}0.944$). A.3 Findings and negative results: why τ and n_{95}/L fail as regime variables; the refuted absolute-support eviction budget; H2 (2-bit score ranking, $\rho = 0.52\text{--}0.77$) and the move to a 3-bit base layer; Gaussian-logit simulation as a qualitatively wrong regime map. A.4 Per-model tables at every context.